PromoFinder: finding promos in microdrama screenplays

Sumit Prabhakar · Kuku AI Generalist Assignment, Problem 1

App: `deltaparticle.github.io/Kuku-Assignment` Code and raw outputs: `github.com/deltaparticle/Kuku-Assignment`

Summary. PromoFinder reads a timestamped screenplay and returns ranked 20–60 s promo clips: a hook in the first 3 s, an unresolved cut, and clear to a stranger. The key decision: *the model may only name script line IDs; code computes every timestamp and enforces every rule.* On the 50-episode pack it returns 30 promos + 2 montages, all passing the hard rules; a blind judge prefers it to a keyword baseline 13–4. It runs on Gemini's free models, and a full show takes under 3 minutes on a paid key.[1]

1 Problem choice and assumptions

I chose Problem 1 because it sits closest to spend: short-drama apps run thousands of ad variants cut from the show itself, and one editor watching every episode does not scale to weekly releases. Its output is also checkable, so quality claims can be tested. **Assumptions:** screenplay times equal video times; promos carry no voice-over or text, so a stranger must follow the footage alone; the audience is US viewers on short-video platforms; a cut never falls inside a spoken line or sound effect; the title twist ("he was a Duke") *sells* the show while villain defeats are spoilers; sexual content (eps 49–50) is flagged as an ad-policy risk.

2 How it works

Each screenplay block becomes a line with an ID (`E07_L042`), a type (camera, action, sound, dialogue) and start/end times. A time is a *safe cut* if no dialogue or sound is playing. The model only ever sees and returns IDs.

1. **Story bible** (1 call): who is who, which twists sell, which payoffs spoil. Spoiler episodes are a 2-of-3 vote.
2. **Candidates** (1 call per episode): three moments as start, hook, cliffhanger and end line IDs.
3. **Validate** (code): hook within 3 s of the start, end on the cliffhanger and before the line that answers it, cuts on safe points, 20–60 s. A failing clip goes back to the model with the legal alternatives code computed.
4. **Score**: a judge rates hook, cliffhanger, clarity, intensity, plot and genre fit (1–5), quoting evidence first and calibrated per batch, giving a weighted composite c (hook and cliffhanger weigh 3, clarity 2, the rest 1). A text *attention* check adds what video tools measure: time to first voice, motion, close-ups, dead air (a).
5. **Rank**: the top 40 are ordered by the judge three times in shuffled order and combined by Borda count b:

$$f = 0.6\,b + 0.4\,c - 0.15\,[\text{spoiler}] - 0.25\,[\text{ad policy}] - 0.1\,\tfrac{e-1}{49} + 0.15\,(a - 0.55) \quad (e = \text{episode})$$

   then a quality bar, one cut per scene, at most two per episode, and diversity across episodes and hook types. *Why these weights:* set by hand with a human in the loop, re-reading the picks after each run until the order matched editorial judgement (the judge's ranking is steadier than its scores, so it weighs 0.6). They are tuned only on Gemini models and not fitted to labelled data; other or stronger models may need different values (all in `config.yaml`; the editor's approve/reject export is the data to re-fit them).
6. **Montages**: the model joins 2–3 moments; the same validator checks them.

Model vs code vs human: the model reads, proposes, judges and ranks; code owns every timestamp and rule; an editor approves or rejects each pick in the web app. It runs from a CLI or entirely in the browser, with any OpenAI-compatible model chosen in `config.yaml` or the app.

3 Results

Official run on the 50-episode pack (3,727 lines, 103 min), with `gemini-3-flash-preview` proposing candidates and `gemini-3.8-flash` judging; a second, fully independent run measures stability, and a keyword-rule baseline is the comparison.

The top of the list is early setup conflict (altar betrayal, public humiliation, a protector stepping in), which is how short-drama apps lead their ads. The blind A/B is significant on decisive pairs (13 of 17, $p = 0.025$). Two changes did most of the work: code no longer edits the model's hook (the first build silently replaced it in 27% of candidates), and low reasoning on the candidate step made each call about 40× faster (172 s to 4 s) without losing the key scenes. **New input:** on 10 minutes of YouTube auto-captions of an official ReelShort episode (no speakers, no visuals), it found 2 strong promos and 1 weak one.

4 Where it picks badly

- **Strict length rule loses moments**: 16 of 36 rejections were too long or short; on the caption test the best opening line was lost this way.
- **Weak endings**: some clips stop on a sad or throwaway line instead of an open question.
- **Noise below the top 5**: the same clip's score varies by 0.5–1 point between runs; top-10 agreement is 6/10.

[1] Expected cost per 50-episode show: nothing on the free tier, though its daily limits spread a full show over a few days; about Rs 50 to run it in one go on a paid key.

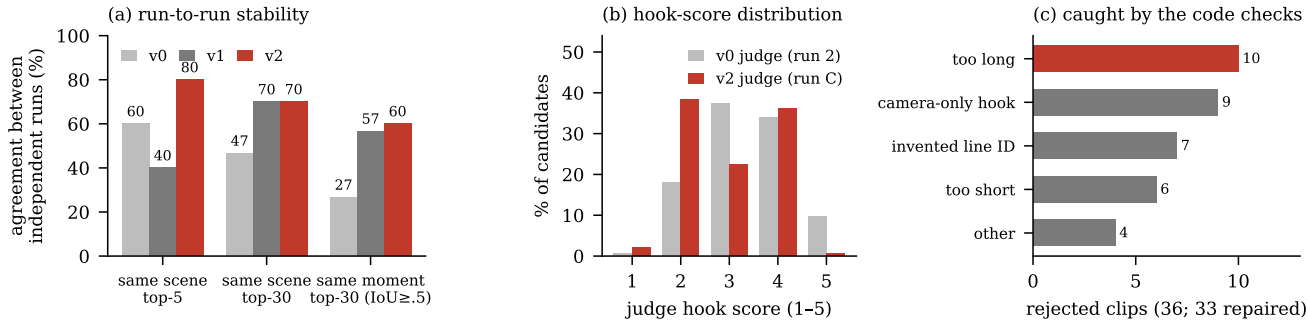


Figure 1: (a) Agreement between two independent runs of each version (v0 first build, v2 final). (b) The early judge scored almost everything 3–5; after calibration the scores separate clips. (c) Clips rejected by the code checks in the official run, by reason; 33 of 36 went back to the model with legal fixes.

Table 1: Official run, 50 episodes.

Valid / rejected candidates	138 / 36
Final list	30 + 2 montages
Hard rules re-checked	32 / 32
Clip length, mean	42 s (22–59)
Blind A/B vs baseline	13–4 (5 ties)
Repeat run, same top-5 scenes	4 / 5
Fast run: calls / time	103 / 2 m 45 s

Table 2: Top of the ranked list (screenplay times, computed by code).

#	In – out	Hook	Ends on
1	E2 00:00–00:45	"I cannot marry you, Juliana."	"Do not disgrace us further."
2	E3 00:58–01:25	"Guards! Remove her!"	". . . let her go."
3	E1 00:11–01:02	"You should turn around, Juliana."	"I changed my mind."
4	E11 00:17–00:51	"Your husband stole my gold watch!"	"Search it."
5	E12 00:44–01:25	"I want to see you crawl, sister."	"Tell me where the garden is."
M1	E2 00:00–00:12 + E3 01:06–01:24 + E4 00:50–01:06: jilted at the altar → a stranger steps in → "It is a vow."		

- **Blind spots**: visual hooks are read from text; the judge and the picker are the same model family; there is no human-labelled ground truth, so nothing here proves a clip will perform.

5 Scale

About 106 model calls per 50-episode show, run in parallel: under 3 minutes instead of 41 (slower on a free key). A company gateway or stronger models plug in through `config.yaml`. An editor reviews about 10 ranked clips instead of watching 50 episodes. The real proof is a live split test of tool- against editor-picked promos on equal spend, compared on hook rate, hold rate, click-through and cost per install. **Next:** re-fit the weights from editor approvals, add audio/visual signals, use a second-family judge, and label a small human gold set.

6 AI tools used, and where they got it wrong

Claude Code harness with Qwen models was used in this project; Gemini models run inside the pipeline. **Where it went wrong:** asked to fix clips that came out too long or short, Qwen wrote a "rescue" that moved the start and *silently replaced the hook* with whatever line came first. All tests passed; only a manual read of the picks caught it (the #1 promo had lost the moment it was chosen for). **The fix** was a single invariant: the hook and cliffhanger are pinned anchors, and any repair may only adjust the clip around them. A one-line check compares the hook text before and after every repair and rejects the clip if it changed, so a silent swap can no longer reach the output.

7 Market analysis: how competitors promote

ReelShort, DramaBox and their peers win viewers mainly through paid short-video ads cut from the shows themselves. We studied their running ads and published industry analyses; each observation set a design rule.

- **Long cuts lead paid feeds.** Most of the competitors' top video ads we reviewed were 10–30-minute compilations, not short clips. A 20–60 s pick therefore also has to work as the cold open of a long ad, which is why the 3-second hook is enforced in code rather than left to the model.
- **Hidden identity sells.** Secret heirs, disguised billionaires and "years later he returns" were the most common premises in competitors' ad copy, alongside revenge, contract marriage and betrayal. So the title twist ("he was a Duke") is treated as a selling point, not a spoiler, and genre fit is scored.
- **Ads open on conflict and come from episode 1.** Long-running drama ads most often open mid-conflict rather than on set-up, and ReelShort's best-known creative was cut from episode 1. Hence camera-only hooks are rejected and early episodes get a small ranking preference.
- **Volume is the bottleneck.** Short-drama apps run thousands of creative variations, and each wears out within days. Hand-picking moments cannot keep that pace, which is the gap this tool fills.